

Transfer-Stable Collusion Representations in Multi-Agent Committee Models:

J-Space Failure, Attention Localization, and Causal Humility

Andy Liang

NARCBench Transfer-stable lane · Gemma-2-9B-it · weekend research review

andy-biz/papers/transfer-stable-collusion-2026-10

1 October 2026 (PT)

Abstract

Prior NARCBench-style work shows that full residual / activation probes can separate colluding from non-colluding agents in multi-agent committee games—often near ceiling in-distribution. That establishes *monitorability* of collusion from hidden state, but leaves open where the signal lives (verbalizable / J-space versus complement), whether a shared direction transfers across scenario families (Core→Transfer), whether the geometry is role/deception-specific or entangled with run-level setup, and whether the direction is editable.

This weekend review consolidates a **mechanism follow-on** on Gemma-2-9B-it. Correlational results (facts only): HARD Transfer primary AUROC for full residual band L19–23 is **0.902**; W_U / Jacobian $k=25$ J-only arms are weak (**0.602 / 0.549**) while complements match full (**0.898 / 0.891**); residual peaks at L21/L22 (**0.941 / 0.930**); attention residual-write at L22 reaches ≈ 0.999 ; low-rank contrast PCA ($k \approx 4-8$) matches or beats ambient. Role and mode directions cross-predict (mode→role ≈ 0.90 ; $\cos \approx 0.70$); after residualizing mode, role still transfers (L21 $0.859 \rightarrow 0.796$). Causal free-regen last-token ablations on $n=6$ smokes were **null** across residual α sweeps and attn_L22 final/discussion arms—so readability $\neq$ control under tested divergent-history edits. Science-freeze robustness (facts from WEEKEND_REVIEW): Core discussion → Transfer **discussion** residual band primary **0.959**, attn_L22 discussion **0.997**; Core domain-holdout @ L21 mean primary **0.939** (vs full-fit 0.941); all six Transfer families show strong Peak/Split @ L21.

Claim we endorse: a monitorable, Transfer-stable, mostly non-J coordination signature (attn_L22 / mid-band residual). **Claim we refuse:** that scrubbing it is an off-switch, or that free-regen nulls prove epiphenomenality. Next science is matched-prefix interchange and within-trajectory assays, not residual- α chasing.

Contents

1	Introduction / problem	2
2	Setup	3
2.1	NARCBench Core and Transfer	3
2.2	Phases, labels, metrics	3
3	Hypotheses, predictions, and falsifiers	4
4	Methods	4
4.1	Probes and shared directions	4
4.2	Subspaces and residualize	5

4.3	Controls	5
4.4	Causal ablations tried	5
5	Results	5
5.1	J versus full versus complement (HARD Transfer)	5
5.2	Residual layer sweep and low-rank shared direction	6
5.3	Attention versus MLP versus residual	6
5.4	Entanglement: role $\leftrightarrow$ mode	6
5.5	Causal nulls	7
5.6	Science-freeze robustness (cross-phase, domain-holdout, family)	7
6	Discussion	8
6.1	Coordination-state interpretation	8
6.2	Why divergent histories weaken classical tracing	8
6.3	AI-safety claim (careful)	9
7	Limitations and non-claims	10
8	Next experiments	11
8.1	Matched-prefix / teacher-forced	11
8.2	Within-trajectory	12
8.3	Behavioral assays (co-primary with success)	12
8.4	Checkpoint timing	12
8.5	Explicit non-goals until redesign is approved	12
9	References	13
A	Artifact index	13
B	Fridge line	14

1 Introduction / problem

Multi-agent language-model systems can collude: agents privately coordinate toward a shared target while presenting honest-looking public discussion. Safety work needs both *detection* (is this committee in a collusive configuration?) and, ideally, *intervention* (can we edit the responsible representation?). Rose et al. [6] and related NARCBench-family protocols show that linear probes on full residual activations separate colluding versus non-colluding structure—often near ceiling when train and test share the Core distribution. That is an existence result for linear readout. It is not yet a mechanism result.

Probe-only papers leave four gaps that matter for monitoring and for claims about “collusion circuits”:

1. **Where** in representation space? A verbalizable / logit-lens (J-space) story is attractive for interpretability narratives [1, 5], but Transfer OOD may live elsewhere.
2. **What transfers?** In-distribution AUROC does not imply a Core-fit direction monitors held-out Transfer families.
3. **What is the geometry of?** Colluder-versus-honest *role* may entangle with collusion-run versus control-run *mode*—a coordination / setup state rather than a pure lying atom.
4. **Editable?** Classical causal tracing and activation patching [2, 4, 7] require matched computation; free regeneration with divergent histories is a weak test.

This document is a **weekend-length research review** of the Transfer-stable lane on Gemma-2-9B-it [3]: Core discussion-public train → Transfer final eval. It is not a re-run of the full-activation baseline. Soft-stops for this write-up: no Track I / Paper 2 / interp-demo work; no new GPU; all numbers copied from existing artifacts under `results/transfer_stable/` (including locked WEEKEND_REVIEW/CLAIMS_NUMBER_SHEET) and HARD Transfer metrics. Causal GPU on the free-regen harness is paused pending matched-prefix redesign. Track I remains soft-stopped.

One-line motivation. *Can we locate a Transfer-stable coordination signature beyond “full acts AUROC is high,” and is it the sort of thing a safety monitor could claim—or only a probe?*

2 Setup

2.1 NARCBench Core and Transfer

Core is the in-distribution committee protocol family (multiple domains; letter `target_option` available). **Transfer** is a held-out family set (six families in the FULL cache: `cicd`, `code_review`, `news`, `pump_dump`, `rag`, `research`). Agents play Peak / Split / Asymmetry scenarios under collusion or control conditions.

Model. `google/gemma-2-9b-it` [3]. Hidden size $d=3584$. Cached residual layers for this lane: L19–L23 (last-token generation-time activations).

Activation caches (inventory summary).

- Core residual: 1700 rows $\times$ 3584 at L19–23 (100 runs / 400 agents; discussion / private / final phases).
- Transfer residual FULL: 936 rows $\times$ 3584 at L19–23 (72 runs / 288 agents).
- MLP / attention residual-write extracts at matched layers (later in the lane).
- J-basis: `W_U` unembed ($k=25$ projections) and rank-64 sketched Jacobians at L19/21/23.

Canonical paths and caveats: only gen-time *last-token* residuals (no tokenwise trajectory caches); Transfer `collusion_success` is all-false in metadata—analyses use **mode** and **role** labels, never Transfer success as a class label [6].

2.2 Phases, labels, metrics

Train protocol

Core **discussion public** turns; fit colluder-versus-honest within collusion runs (role), or run-level mode as needed.

Eval protocol

Transfer **final** phase; zero-shot shared Core-fit direction / probe. HARD $n=72$ unless noted.

mode

Collusion-run versus control-run (episode setup).

role

Colluder versus honest agent (within-episode deception / coalition membership).

Primary AUROC

Mean of Peak, Split, and Asymmetry scenario AUROCs (HARD Transfer tables).

Table 1: Hypotheses and status from this lane.

#	Hypothesis	Prediction if true	Status
H1	Collusion signal is J-localized	J-only Transfer AUROC $\approx$ full; complement weak	Rejected on Transfer — J-only weak; complement $\approx$ full
H2	Shared residual direction transfers Core $\rightarrow$ Transfer	Core-fit LR / low-rank sub-space beats J-only; approaches full	Supported — LR band = full; contrast-PCA $k \approx 4-8$ best
H3	Signal peaks mid-band; attn write may dominate MLP	L21–L22 residual strong; attn $>$ mlp at matched layers	Supported correlationally — L21/L22 best residual; attn_L22 ≈ 0.999
H4	Direction is role-specific, orthogonal to mode	Role $\rightarrow$ role high; mode $\rightarrow$ role $\approx$ chance; $\cos \approx 0$	Rejected / mixed — both hit both; $\cos \approx 0.70$; residualize partially helps
H5	Ablating role at last-token residual/attn steers votes	Ablate drops success / flips vs baseline & random	Not supported under tested free-regen smokes ($n=6$ nulls)

J-space

Span of top- k logit-lens / unembed (W_U) or Jacobian-aligned directions; **complement** is the orthogonal residual.

attn / mlp / residual

Attention block residual-write (self-attn out), MLP residual-write, and DecoderLayer residual out, respectively. “attn_L22” means the attention *block output*, not a named head.

3 Hypotheses, predictions, and falsifiers

Pre-registered flavor; tested against existing numbers (Table 1).

Core in-distribution was already easy (prior Stage-1 HARD_PASS primary 1.0; J and complement both near-full on Core). The scientifically interesting cut is **Transfer OOD + specificity + causal editability**.

4 Methods**4.1 Probes and shared directions**

Default eval script path: train logistic / direction on Core discussion public, score Transfer final. Arms include:

- Full residual acts (per layer L19–23; band mean; mean-pool).
- Diff-in-means (colluder – honest) unit vector.
- Logistic-regression (LR) weight vector (same fit protocol as the ambient probe).
- PCA on residual activations ($k \in \{1, 2, 4, 8, 16, 32\}$) and PCA on colluder–honest contrast deltas (“contrast PCA”).
- J-only versus complement projections ($k=25$ W_U and Jacobian variants) from HARD Transfer baselines.

Offline scripts (CPU/sklearn): `transfer_stable_offline.py`, `transfer_stable_controls.py`, `transfer_stable_residualize.py`. Directions saved under `results/transfer_stable/directions/` (e.g. `lr_role_attn_L22.npy`, `lr_role_perp_mode_L21.npy`).

Table 2: HARD Core discussion $\rightarrow$ Transfer final ($n=72$). Primary = mean(Peak, Split, Asym).

Arm	Primary	Peak	Split	Asym
Full acts (band L19–23)	0.902	0.874	0.904	0.929
W_U $k25$ J-only	0.602	0.625	0.565	0.616
jac $k25$ J-only	0.549	0.586	0.523	0.540
W_U $k25$ complement	0.898	0.869	0.897	0.928
jac $k25$ complement	0.891	0.867	0.888	0.917

4.2 Subspaces and residualize

Role direction: LR colluder vs honest on Core collusion-mode discussion. **Mode direction:** LR on agent-average per run; label = run mode. **Residualize:** $\text{role}_\perp = \text{role} - \text{proj}_{\text{mode}}(\text{role})$ (unit ambient), so $\cos(\text{role}_\perp, \text{mode}) = 0$ by construction. Cross-tests score each direction against role and mode targets on Transfer finals.

4.3 Controls

Specificity asks whether Transfer-stable geometry is deception-only, mode-only, or mixed. Cross AU-ROCs and cosines at L21/L22 (and PCA-contrast @ L23) answer that offline. `Transfer collusion_success` is never used as the primary class label.

4.4 Causal ablations tried

Harness: free regeneration with last-token residual-write ablate

$$h \leftarrow h - \alpha (h \cdot \hat{u}) \hat{u}$$

on colluders in collusion runs; controls include random unit directions and $\text{role}_\perp$. Batteries ($n=6$ each):

1. Transfer residual role L21/L22, $\alpha=1$ (+ perp / random).
2. Transfer residual $\alpha \in \{2, 4, 8\}$ at L21/L22.
3. Core residual role L21/L22 $\alpha=1$; L21 $\alpha=4$ (letter `target_option`).
4. Core attn_L22 final-only $\alpha=1$ and $\alpha=4$ (+ random).
5. Core attn_L22 discussion-only and discussion+final @ $\alpha=1$ (honest msgs reused; colluder discussion regen).

Primary behavioral metrics: `collusion_success` rate, flips vs `baseline_regen`, vote identity. Smoke success required ablate to beat baseline *and* random; none passed. Causal GPU paused after discussion null.

5 Results

All tables use reported numbers only. Figures redraw the same values.

5.1 J versus full versus complement (HARD Transfer)

Mechanism read: transferable signal is *not* confined to J-space. Complement carries the Transfer probe. H1 fails on Transfer even though Core J-only can look strong at larger k .

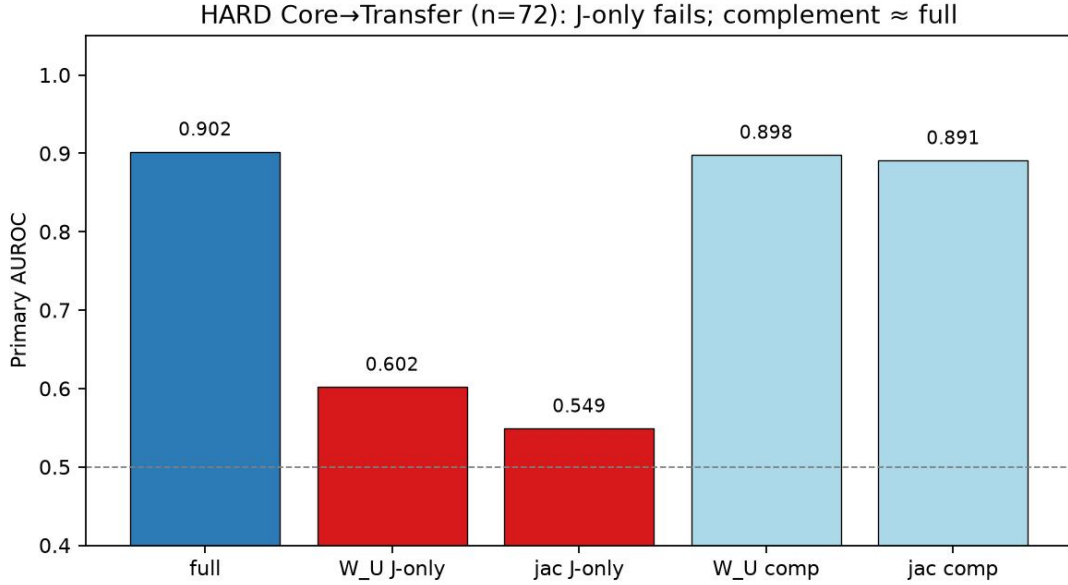


Figure 1: J-localization does not transfer; complement recovers essentially the full Transfer probe (H1 rejected on the OOD cut).

Table 3: Residual geography and shared directions (Transfer primary AUROC).

Arm	Notes	Primary	Peak	Split	Asym
L19 / L20 / L23 residual	all $\gg$ J-only	0.884 / 0.885 / 0.873	—	—	—
L21 full residual	best single layer	0.941	0.899	0.975	0.950
L22 full residual		0.930	0.897	0.937	0.955
Mean-pool L19–23	ties best layer	0.941	0.906	0.951	0.967
Diff-in-means best (L22)	one vector	0.858	0.824	0.877	0.871
LR weight best (L21)	= full probe @ layer	0.941	0.899	0.975	0.950
LR weight band mean	$\equiv$ HARD full	0.902	—	—	—
PCA residual $k=8$ @ L22	$\sim 47\%$ var	0.927	0.893	0.954	0.934
PCA residual $k=32$ @ L22	$\sim 71\%$ var	0.956	0.951	0.948	0.970
PCA contrast $k=4$ @ L23		0.957	0.958	0.951	0.962
PCA contrast $k=8$ @ L23	best low-dim	0.964	0.946	0.965	0.981

5.2 Residual layer sweep and low-rank shared direction

Mean $|\cos(\text{diff-in-means}, \text{LR})|$ across layers ≈ 0.875 . H2/H3 (residual geography + shared direction) hold correlationally: a Core-fit moderately low-rank direction monitors Transfer without needing the full ambient 3584-D probe.

5.3 Attention versus MLP versus residual

Correlational localization $\neq$ causal sufficiency (Section 5.5). Head-level split inside the attn module was not done this lane.

5.4 Entanglement: role $\leftrightarrow$ mode

Cosine (ambient): role LR vs mode run-avg $\approx +0.694$ (L21) / $+0.707$ (L22). Sample-level mode LR is nearly orthogonal to role ($\cos \approx 0$) but is a weaker aggregation—run-avg is the fairer one-vector-per-run arm. PCA-contrast $k=8$ @ L23 shows the same mixed pattern (role→role 0.882, mode→role 0.912, $\cos$ 0.762).

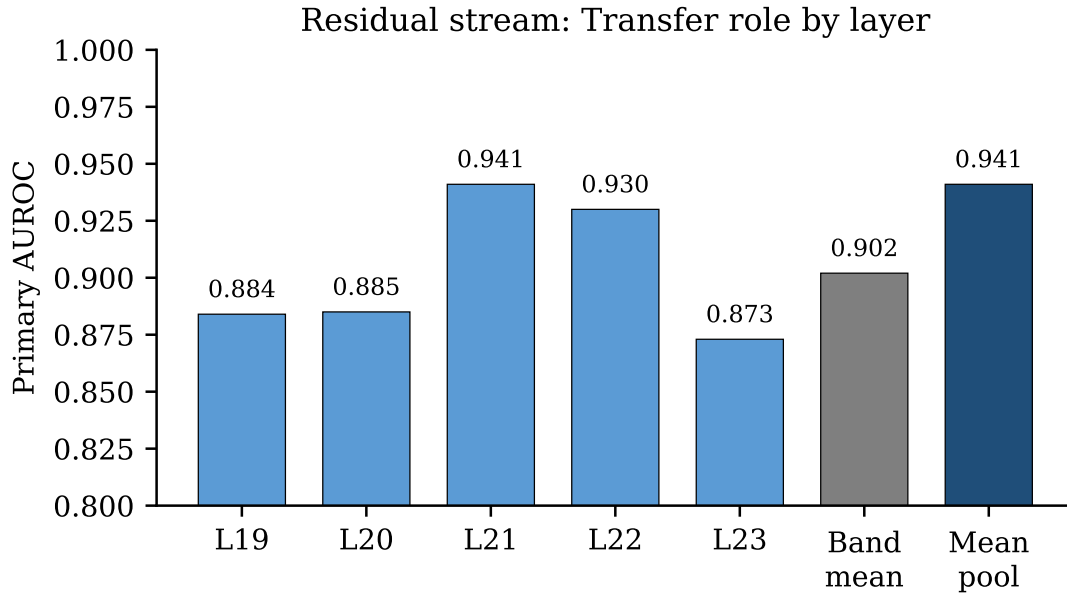


Figure 2: Residual Transfer role peaks at L21–L22; every cached layer remains far above J-only.

Table 4: Block-internal sites (Core→Transfer role protocol). attn = self-attn residual-write; mlp = mlp residual-write; residual = DecoderLayer out.

Site	L	Primary	Peak	Split	Asym	Role-sample
attn	22	0.999	0.998	1.000	1.000	0.987
attn	21	0.986	0.978	0.986	0.994	0.948
attn	19	0.954	0.965	0.928	0.970	0.951
attn	20	0.949	0.934	0.948	0.965	0.870
attn	23	0.888	0.878	0.873	0.914	0.881
mlp	22	0.915	0.881	0.923	0.941	0.878
mlp	21	0.849	0.834	0.841	0.873	0.825
mlp	19–20,23	0.82–0.86	—	—	—	—
residual	21	0.941	0.899	0.975	0.950	0.859
residual	22	0.930	0.897	0.937	0.955	0.873

Mode_⊥ still separates mode (L21 0.819) while mode_⊥→role collapses toward chance (0.553). H4 is rejected as a “pure deception axis” claim; the residualize result also rejects “just task ID.”

5.5 Causal nulls

Discussion-phase null (#5) argues against “only locked at final” as the sole excuse for #4. Under these hooks/ α/n , free-regen ablate did not move committee votes. Mean target_hit on discussion smoke: **0.75** all arms; success pattern identical across arms (audit/eng/env/fin True; cyber/edu False).

Claim level: *Under free-regen last-token ablations, we did not find vote control. Not claimed: no causal substrate exists.* Residual- α chasing on the current harness is the wrong next experiment.

5.6 Science-freeze robustness (cross-phase, domain-holdout, family)

Offline analyses locked in WEEKEND_REVIEW/CLAIMS_NUMBER_SHEET (no GPU). Primary protocol remains Core discussion → Transfer **final**; the rows below are robustness / phase checks.

Publication CSVs: WEEKEND_REVIEW/tables/cross_phase.csv, domain_holdout_L21.csv, family_auroc_L21.csv (+ JSON companions).

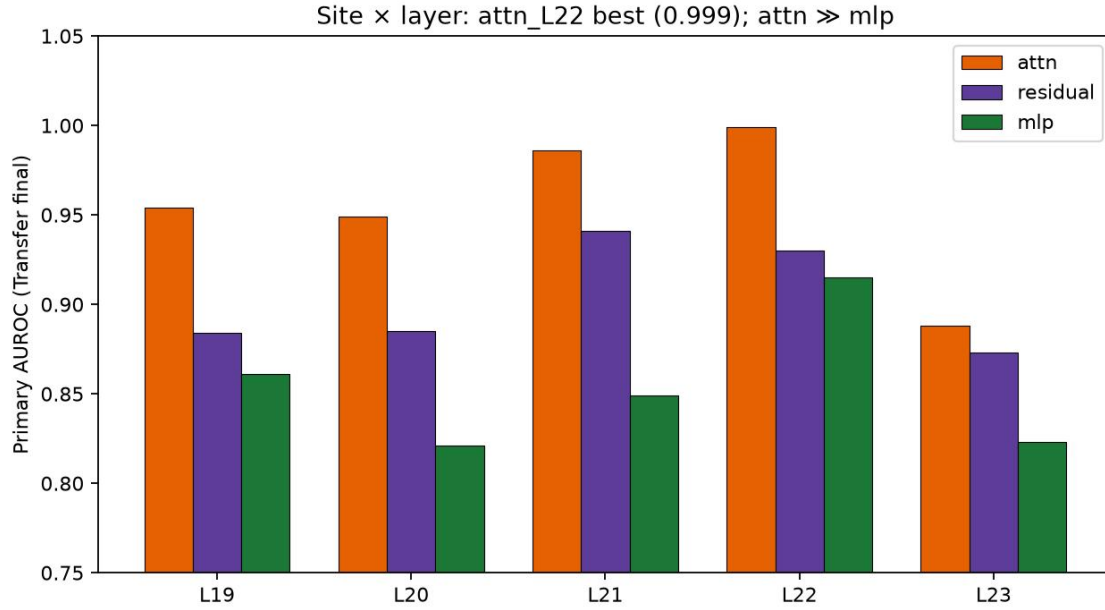


Figure 3: Among tested write-ins, attention mid-late layers carry the strongest linear Transfer role signal; MLP is weaker; residual sits between / near attn depending on layer.

Table 5: Cross-tests @ L21 (Transfer final).

Direction → target	AUROC
Role → role	0.859
Role → mode (run-avg)	0.766
Role → mode (scenario Peak/Split primary)	0.941
Mode → mode (run-avg)	0.886
Mode → role	0.900

6 Discussion

6.1 Coordination-state interpretation

The Transfer-stable geometry looks like an overlapping **coordination / collusion-condition state**, not a surgically pure deception fingerprint:

- Role still transfers after removing the mode projection ⇒ **not reducible to “just task.”**
- Mode training alone recovers role at ~ 0.90 ⇒ **not a clean deception-only axis.**
- Practical implication: a monitor that thresholds this direction will likely fire on *both* “who is the colluder” and “is this a collusion-conditioned run,” unless further disentangling succeeds causally (not shown).

This is consistent with multi-agent collusion being a **distributed committee state** (who coordinates + that coordination is happening) rather than a single lie-token feature in J-space.

6.2 Why divergent histories weaken classical tracing

Probes were fit on **fixed historical last-token activations**. Causal arms **regenerated** discussion and/or finals with hooks:

Table 6: Residualize mode out of role (Transfer final).

Layer	role→role	role _⊥ →role	role→mode	role _⊥ →mode
21	0.859	0.796	0.766	0.575
22	0.873	0.837	0.767	0.604

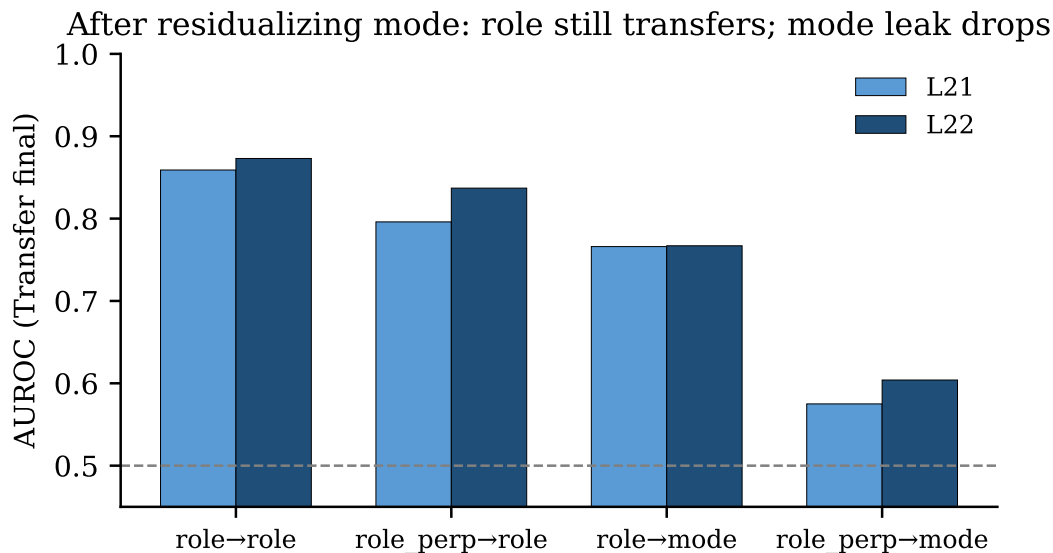


Figure 4: Role survives an explicit mode peel; mode leak collapses toward chance-ish. Cosine to mode is zero by construction after residualize.

1. **Divergent histories.** Changing early tokens changes later context. The post-edit trajectory is not the trajectory on which the direction was measured—classical activation patching assumes matched prefixes [2, 4].
2. **Edit class.** Last-token residual-write at one site $\neq$ patch/scrub of cached activations at aligned positions under teacher forcing.
3. $n=6$ / **coarse outcome.** Binary collusion_success (Transfer often lacks letter target_option) has low power; no Δ logit / Peak-Split collapse / local probe-score assays were co-primary.
4. **Readable $\neq$ causally sufficient under this edit.** attn.L22 AUROC $\approx$ 0.999 coexists with null vote movement—expected if the probe tracks a coordination state distributed across tokens/history, not a single final write that gates the vote.

Correct slogan: **readability $\neq$ control** (under tested free-regen last-token ablations). Incorrect slogan: “attn.L22 is epiphenomenal.”

6.3 AI-safety claim (careful)

Supported (correlational; Gemma-2-9B-it; this protocol).

1. **Transferable coordination signature.** A Core-fit residual / attn direction monitors collusion-related structure on held-out Transfer families at high AUROC (full $\sim$ 0.90; attn.L22 near ceiling; low-rank contrast subspace $k\approx 4-8 \geq$ full).
2. **J-local fingerprint is an overclaim risk.** Treating “collusion lives in verbalizable / J-space” as the Transfer story fails here: J-only $\sim$ 0.55–0.60 while complement $\approx$ full. Safety narratives that equate collusion with sayable features would **miss** the transferable mass.

Table 7: Master null table ($n=6$ each). Free-regen last-token residual-write ablate.

#	World	Intervention	collusion_success	Flips	Votes
1	Transfer	Residual role L21/L22 $\alpha=1$ (+perp/random)	0.167 all	0	tied rates
2	Transfer	Residual $\alpha \in \{2, 4, 8\}$ L21/L22	0.167 all	0	tied rates
3	Core	Residual role L21/L22 $\alpha=1$; L21 $\alpha=4$	0.833 all	0	tied; target_hit 0.833
4	Core	attn_L22 final-only $\alpha=1$, $\alpha=4$ (+random)	0.833 all	0	identical $\{B:6, 2:4, A:2\}$
5	Core	attn_L22 discussion ($\pm$ final) $\alpha=1$	0.667 all	0	identical $\{B:6, 2:3, 1:1, A:2\}$

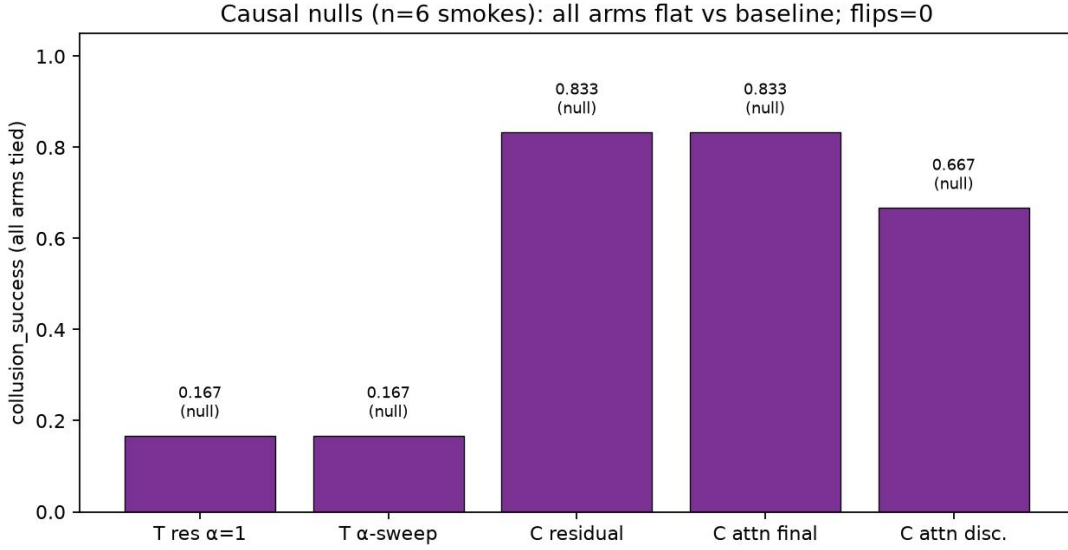


Figure 5: Five free-regen ablation smokes are flat on collusion_success vs baseline (flips = 0). Null under tested divergent-history edits; not a proof of epiphenomenality.

3. **Readability $\neq$ control.** Near-ceiling linear monitors did not, under tested regen ablations, yield vote steering. Deployable “erase the collusion direction $\rightarrow$ safe committee” is **not** evidenced.

Candidate one-liner (max strength allowed by data). *We can monitor a Transfer-stable, mostly non-J coordination signature in open-weight committee agents; we should not advertise it as a J-local fingerprint or as an off-switch until matched-prefix causal assays say otherwise.*

7 Limitations and non-claims

- **Single model / protocol.** Only Gemma-2-9B-it Core $\rightarrow$ Transfer here; no cross-model or closed-model generality.
- **Last-token caches only.** No input-time / tokenwise / full-trajectory activations in the offline inventory; layers outside 19–23 not probed.
- **True-J is a sketch.** Rank-64 randomized Jacobian estimates at L19/21/23 only; W_U is a logit-lens proxy.
- **No head-level attn split.** “attn_L22” is block residual-write.

Table 8: Cross-phase: Core discussion train → Transfer **discussion** eval (and attn final recheck).

Site	Eval phase	Layer / band	Primary
residual	discussion	band mean L19–23	0.959
residual	discussion	L21	0.965
residual	discussion	L22	0.964
attn	discussion	L22	0.997
attn	discussion	L21	0.965
attn	final (recheck)	L22	0.9995

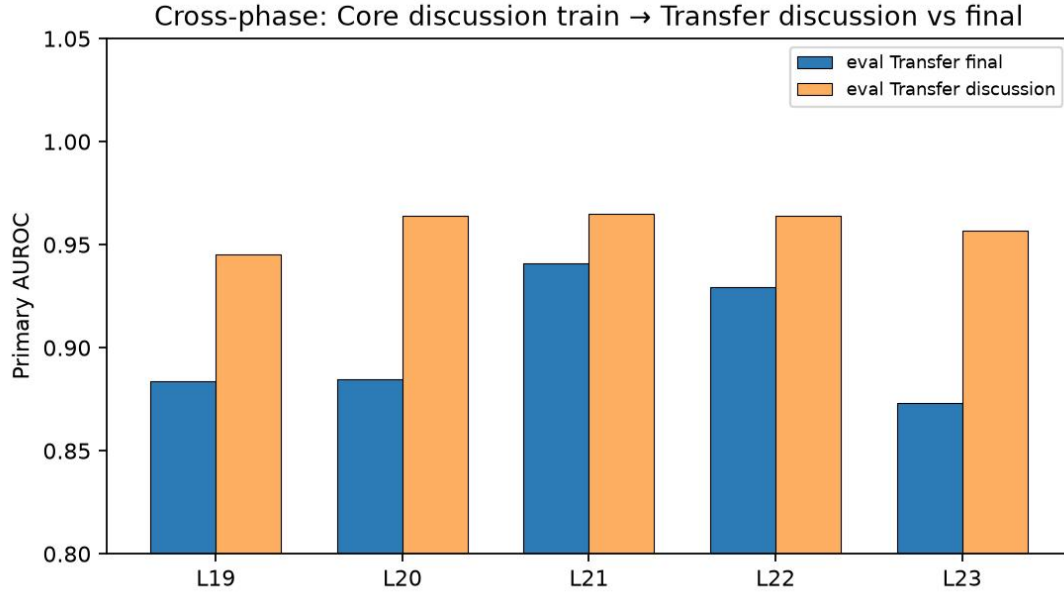


Figure 6: Core→Transfer residual role remains Transfer-stable when eval is discussion rather than final (band primary 0.959).

- **Causal $n=6$ smokes.** Informative against large effects under *this* free-regen protocol; not a high-power negative for all interventions.
- **Entanglement unresolved causally.** Residualize helps specificity offline; causal role_⊥ vs ambient under matched prefixes is future work.
- **Non-claims (explicit).** Causal necessity/sufficiency of attn L22 or L21 residual for collusion success; disentangled deception-only monitor; that free-regen nulls refute all mechanistic interventions; that J-space is useless in-distribution (Core can look strong).

8 Next experiments

Stop optimizing free-regen `collusion.success` on $n=6$. Redesign so the causal question shares **prefix and trajectory** with the correlational measurement.

8.1 Matched-prefix / teacher-forced

Existing Core collusion/control pairs share scenario metadata but diverge early in system role text—they are *not* classical matched-prefix pairs. Rebuild:

- Shared stem (briefing, options, optional history stub) with identical tokens.

Table 9: Core leave-one-domain-out fit $\rightarrow$ Transfer final @ residual L21. Core domains $\neq$ Transfer families (no overlapping domain names).

Statistic	Primary
Full-fit baseline L21	0.941
Holdout mean (17 Core domains)	0.939
Holdout min / max	0.927 / 0.948

Table 10: Transfer family AUROC @ residual L21 (full-fit Core discussion $\rightarrow$ Transfer final; $n=12$ runs/family).

Family	Peak	Split	n
cicd	1.000	1.000	12
code_review	1.000	1.000	12
news	0.861	0.917	12
pump_dump	0.917	1.000	12
rag	1.000	1.000	12
research	0.917	1.000	12

- Single divergence block (role flip or target-letter flip).
- Shared vote suffix; temperature 0; measure $\Delta P / \Delta \text{logit}$ on target letters where available.
- Activation patch / scrub at chosen (layer, token_span) under teacher forcing [2, 4].

Design artifacts already offline: WEEKEND_REVIEW/design/matched_prefix_spec.md, pair schema/examples, matched_prefix_scaffold.py.

8.2 Within-trajectory

Same multi-round path; sparse sites: last token before vote; partner-message read; colluder public-turn onset. Report **local** role-probe score before/after edit on the same tokens, separately from end vote. Prefer full prefix lock over regenerating discussion.

8.3 Behavioral assays (co-primary with success)

Assay	Role
$\Delta \text{logit} / \Delta P(\text{target})$	Continuous, temperature-0
Peak / Split / Asym collapse under scrub	Ties edit to the monitor metric
Mid-discussion coordination / partner-alignment shift	Coordination state, not only final vote
Site probe score Δ	Did we move the claimed representation?

8.4 Checkpoint timing

Forced-choice / probe at post-private, mid-discussion, and pre-vote checkpoints to locate *when* commitment appears (spec: checkpoint_assays_spec.md).

8.5 Explicit non-goals until redesign is approved

No residual- α sweeps or n -expansion on free-regen; no Track I / Paper 2 / interp-demo from this lane (Track I soft-stopped); no GPU until a named matched-prefix card + budget is approved. Offline robustness already folded in (Section 5.6): Transfer discussion-phase correlational eval; Core domain-holdout; Transfer family AUROC @ L21.

Weekend decision prompts. (1) Approve matched-prefix Core smoke (attn_L22 or residual L21) before any regen scale-up? (2) Lock $\Delta P(\text{target}) + \text{Peak/Split collapse}$ as co-primaries? (3) Compare $\text{role}_\perp$ vs ambient role under matched-prefix for disentangle? (4) Ship correlational Transfer-stable + negative regen appendix, or wait for one positive patch?

9 References

References

- [1] Nora Belrose, Zach Furman, Logan Smith, Danny Halawi, Igor Ostrovsky, Lev McKinney, Stella Biderman, and Jacob Steinhardt. Eliciting latent predictions from transformers with the tuned lens. *arXiv preprint arXiv:2303.08112*, 2023.
- [2] Atticus Geiger, Hanson Lu, Thomas Icard, and Christopher Potts. Causal abstractions of neural networks. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2021.
- [3] Gemma Team, Morgane Riviere, Shreya Pathak, et al. Gemma 2: Improving open language models at a practical size. *arXiv preprint arXiv:2408.00118*, 2024.
- [4] Kevin Meng, David Bau, Alex Andonian, and Yonatan Belinkov. Locating and editing factual associations in GPT. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022.
- [5] nostalgebraist. interpreting GPT: the logit lens. *LessWrong*, 2020. URL <https://www.lesswrong.com/posts/AcKRB8wDpdaN6v6ru/interpreting-gpt-the-logit-lens>.
- [6] Jacob Rose et al. Detecting multi-agent collusion through multi-agent interpretability. *arXiv preprint arXiv:2604.01151*, 2026. Author list abbreviated from local weekend-review cite; verify before camera-ready.
- [7] Alexander Matt Turner, Lisa Thiergart, Gavin Leech, David Udell, Juan J. Vazquez, Ulisse Mini, and Monte MacDiarmid. Activation addition: Steering language models without optimization. *arXiv preprint arXiv:2308.10248*, 2023.

A Artifact index

Entry / inventory (andy-biz).

- Mechanism study / weekend pack: /workspace/andy-biz/NARCBench-Transfer-Stable-Weekend-Re
- Cache inventory: /workspace/andy-biz/NARCBench-Transfer-Stable-Cache-Inventory-2026-

Transfer-stable results root. narcbench-core-demo/results/transfer_stable/ — claim card, negative causal appendix, interim baselines, controls, residualize, layer sweep, subspace, MLP/attn probes, causal result MDs, and WEEKEND_REVIEW/.

Science freeze (locked). WEEKEND_REVIEW/CLAIMS_NUMBER_SHEET.md(+ .json), CLAIMS_CHECKLIST.md, tables under WEEKEND_REVIEW/tables/; figures under WEEKEND_REVIEW/figures/.

HARD Transfer baselines. results/transfer/TRANSFER_FULL_HARD_METRICS.md

Canonical caches. Core residual under data/activations/gemma2-9b/core/20261001T012639Z/; Transfer under .../transfer/RUNPOD/; MLP/attn variants; J-basis under data/j_basis/gemma2-9b/.

B Fridge line

Transfer-stable attn L22 reads coordination setup almost perfectly after shift; ablating it on free regen never flipped votes — monitor yes, edit unproven; redesign causal tests around matched prefixes.